

IAN ABIEL WANGSA

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EDUCATION

National University of Singapore (NUS)

Singapore

B.Sc. in Computer Science (Minor in Economics & Statistics) | GPA: 4.50/5.00

Aug 2025 – Aug 2029

- **Coursework:** Programming Methodology I & II, Data Structures & Algorithms, Linear Algebra, Discrete Structures, Calculus.

Eunoia Junior College

Singapore

GCE A-Levels (Science Stream: H2 Math, Physics, Chemistry, Economics)

Jan 2021 – Nov 2022

- **Academic Excellence:** Awarded Dean's List (2021, 2022). Received Edusave Merit and Academic Achievement Awards.
- **Varsity Softball:** 2nd Place in National School Games (2022); Awarded National Colours Award.

EXPERIENCE

Accenture

Singapore

Summer Intern – Application Support Engineer, Technology (Health & Public Services)

Apr 2026 – Jul 2026

- Engineered an enterprise **Root Cause Analysis (RCA) Generator** leveraging Agentic AI, hybrid retrieval pipelines (BGE chunking, BM25, vectorized NLP), and Azure DevOps integrations to analyze system codebases and log traces directly.
- Leveraged a custom MCP Server alongside automated Playwright authenticators to dynamically scrape SharePoint tokens, powering a Flask web frontend for seamless developer query execution.
- Architected an automated **Test Script Generator** utilizing Claude CLI Agentic AI workflow, custom sub-agents, and Python libraries; seamlessly streamlined incident response and testing workflows from multiple days to **under 30 minutes**, achieving a **95%+ boost in operational efficiency**.
- Remediated critical runtime application bottlenecks, race conditions, and production bugs across enterprise software architectures, ensuring SLA compliance and high availability.

CTES Consulting

Singapore

Software Engineer - Off-Cycle Intern

Jul 2026 – Present

- Engineered full-stack profiling and data extraction pipelines utilizing React, TypeScript, Tailwind CSS, Node.js, and PostgreSQL, deployed on AWS behind Cloudflare DNS/Security layers.
- Developed automated Web Scraping and Session Management modules using Python, Playwright, and Chromium to handle FedAuth cookie authentication and complex portal navigation.
- Built document parsing and report generation engines using `python-docx`, `pypdf`, and `openpyxl` to process candidate profiles and structured document outputs.

D+A Strategies

Remote

Quant Trainee - Apprenticeship

Mar 2026 – Present

- Design, backtest, and stress-test systematic algorithmic trading strategies by using vectorized mathematical pipelines and evaluating statistical arbitrage behaviors over unstructured financial time-series data.
- Construct quantitative modeling environments in Python to evaluate alternative pricing patterns and optimize output.

Republic of Singapore Air Force (RSAF) & Singapore Army

Singapore

Air Force Signals Commander & Specialist Cadet (NSF)

Mar 2023 – Jan 2025

- **Key Deployments:** Selected for Exercise Wallaby (Rockhampton, Australia) to establish joint SAF-RAAF radio communications; participated in SAF Day Parade (Jul 2024).
- **Unit Leadership:** Served as Comms Store IC, Gym Innovation IC, and National Education IC, leading the 160 SQN unit to win the NE CDF Award for two consecutive years (2023–2025).

PROJECTS & HACKATHONS

BrainHack TIL-AI 2026 — DSTA Singapore | 1st Place (Team Pandamonium)

May 2026 – Jun 2026

- **Tech Stack:** Python, Docker, FastAPI, PyTorch, NeMo, YOLOv11, BM25, Transformers, SB3
- Built a finals-ready multimodal AI system with 5 Dockerised FastAPI microservices deployed in an air-gapped environment.
- **ASR:** Fine-tuned NVIDIA Parakeet TDT with custom decoder, FP16 inference, sorted batching, and phrase correction.
- **CV:** Engineered a YOLOv11 target detection pipeline for aircraft, vehicles, and vessels with threshold tuning.
- **Noise:** Built a YOLO-surrogate PGD adversarial noise generator constrained by SSIM/RMSE budgets to fool opponent detectors.
- **NLP:** Deployed a BM25-first RAG pipeline with alternative neural reranking layers and quantized LLM generation.
- **AE:** Developed a MaskablePPO RL navigation agent featuring action masking, reward shaping, and fallback BFS/A* pathing.

NTU BeyondBinary / SunnyHacks 2026 | 2nd Place | React, TS, MediaPipe, Gemini

Feb 2026

- Engineered **SignMeUp** and **Aurora**, an on-device AI accessibility and edge-vision wellness suite utilizing MediaPipe/TFLite for real-time fall detection, hydration tracking, and empathetic eldercare support.
- Integrated **PathGuardian 2.0** voice-first navigation with caregiver alerts alongside a localized, custom LSTM network optimized for high-performance temporal sign language translation (ASL/BSL/SSL).

Rockathon 2026 – By BugleRock | Top 5 | Python, React, PostgreSQL, Claude API

Mar 2026

- Built **DealMachine**, a comprehensive AI-powered platform addressing core challenges in investment banking workflows, significantly reducing manual overhead and accelerating high-quality, compliance-ready document output.
- Developed **Supersearch**, an intelligent assistant using pandas & statsmodels to aggregate live market data for financial analysis.
- Implemented a **Doc Generator** for automated drafting of Information Memorandums and NDAs, alongside a real-time **Deal Dashboard** using Chart.js to provide cross-functional teams with pipeline visibility and analytics.

LEADERSHIP & CO-CURRICULAR ACTIVITIES

NUS Games Development Group | Software Engineer (Pandamonium)

Jan 2026 – Present

- Leading technical development for *Pandamonium*, an interactive 2D murder mystery game developed using C# and Unity.
- Designed a custom procedural board rendering pipeline from scratch and architected centralized state management systems to track game narrative progression and evidence logs.

NUS Sports Club | Executive Member (RunNUS & SunNUS)

Aug 2025 – Present

- Managed event operational logistics, vendor procurement, and budget allocation for major flagship initiatives, including a flagship charity run with over 1,000 active participants.
- Coordinated cross-functional teams to execute large-scale beach events, ensuring strict safety standards and seamless crowd control protocols.

NUSSU VENTures | Operations Executive

Aug 2025 – Present

- Oversee event planning workflows, venue synchronization, and resource management for operations across campus.
- Streamlined operational communications between campus stakeholders, venue managers, and internal student committees.

Project BHID | Programmes Head

Aug 2025 – Present

- Spearheaded curriculum development tailored for adults and children with intellectual disabilities, focusing on engaging cognitive exercises and practical life skills.
- Direct a team of student volunteers, conducting training sessions to ensure effective communication and empathetic facilitation during workshops.

Youth Corps Singapore & Rotaract Club | Volunteer Educator

Jan 2025 – Present

- Taught foundational Scratch programming, logic building, and introductory Artificial Intelligence/Machine Learning concepts to primary school students.
- Mentored young learners through hands-on coding exercises, building early interest in computational thinking and STEM.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript/TypeScript, SQL, C#, HTML, CSS

Frameworks: React, Next.js, FastAPI, Node.js, Spring Boot, Flask, Unity, Tailwind CSS, PyTorch, TensorFlow, Scikit-learn, Pandas

Developer Tools: Docker, AWS, Cloudflare, Git, Linux, Azure DevOps, Power Automate, Playwright, MediaPipe, ChromaDB

Certifications: Anthropic AI Fluency, Anthropic Claude 101, Databricks AI/BI, AWS Cloud Practitioner (CLF-C02), GitHub Foundations, Docker Foundations